



CPSC 100

Computational Thinking

Algorithm, Classifiers and Trees!

Instructor: Firas Moosvi
Department of Computer Science
University of British Columbia

Agenda

- Course Admin
- Debrief on Moore's Law
- Recap Classification
- Decision Trees

Course Admin



Course Admin

- **Learning Log 2** will be posted at 5 PM (after class!)
 - Due at 6 PM today, but has a 48-hr no-penalty grace period
- **Test 2:** Reminder to schedule Test 2 on PrairieTest!
 - Only about half of you have scheduled!
- **Next week:** I will be away for 3B, 3C, 4A, 4B for a conference in Singapore (sorry!)
 - Parsa Rajabi (former CPSC 100 instructor) will be here!

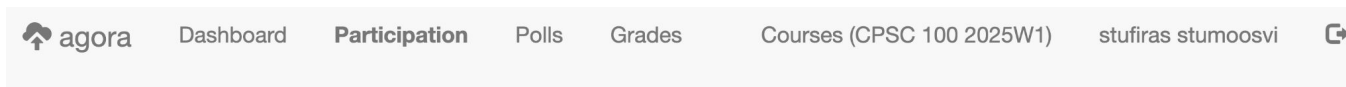
Debrief on Moore's Law



Agora: Discussions and Polling

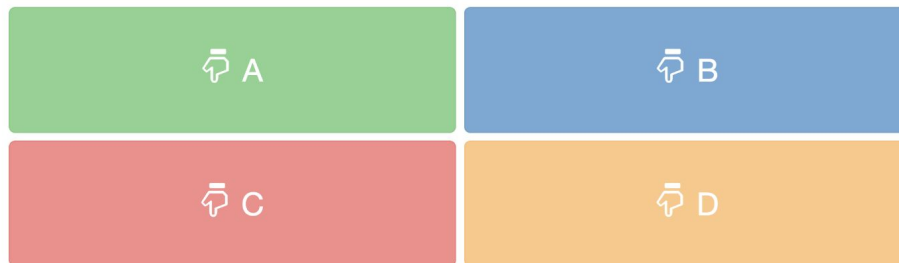
Page: agora.students.cs.ubc.ca (login with your CWL)

Enroll code: **conformitygate**



CPSC 100 2025W1

Participation Points: + (Phantom)



Message Board

Lecture not started.



Participation Question

Is Moore's Law still valid in 2025?

Yes; explain

No; explain

-

-



Slides for Pre-reading

**Slides with yellow borders
will not be covered in
class, but is still testable
content - you should
review this before class.**

Steps to do Classification

Step 1: Start with the data you have

Applicant	Annual Income	Loan Approved?
#1	26 000	No
#2	60 000	Yes
#3	50 000	Yes
#4	47 000	No
#5	12 000	No
#6	108 000	Yes

Step 2: Split data into training and test sets

We chose a 50/50 split for our demo but you could do other splits like 60/40, 70/30. For large datasets, 80/20 split is used

Applicant	Annual Income	Loan Approved?
#1	26 000	No
#2	60 000	Yes
#3	50 000	Yes

} **Training Data**

Applicant	Annual Income	Loan Approved?
#4	47 000	No
#5	12 000	No
#6	108 000	Yes

} **Test Data**

Step 3: Build classifier

(i.e., Find pattern in training set)

Given your training data, can you find a pattern that can tell you when to approve a loan?

Earlier, we decided an annual income of ~\$50,000 seemed like a good cut off point. **That was a classifier!**

Applicant	Annual Income	Loan Approved?
#1	26 000	No
#2	60 000	Yes
#3	50 000	Yes

} Training Data

Step 4: Use classifier on test data

Applicant	Annual Income	Loan Approved?
#4	47 000	?
#5	12 000	?
#6	108 000	?

} **Test Data**

After you come up with a classifier that seems to do okay with your training data, you use it on your test data to see what kinds of decisions it makes.

Step 5: Calculate Accuracy

Applicant	Annual Income	Loan Approved?	Classifier said to...
#4	47 000	No	No
#5	12 000	No	No
#6	108 000	Yes	Yes

} **Test Data**

If the results of your classifier match up with the decisions you've made in your test data, it's looking good.

You can start trying to use it on data that you haven't made any decisions on yet.

Seems Straightforward

- What happens when we have more than one attribute?
- In the example before, we only had to consider annual income
- But what would happen if we had multiple attributes, like 5 or 10 or 100?
- **How do we decide which attribute to use?**

Seems Straightforward

- What happens when we have more than one attribute?
- In the example before, we only had to consider annual income
- But what would happen if we had multiple attributes, like 5 or 10 or 100?
- **How do we decide which attribute to use?**

Decision Trees!

Seems Straightforward

- What happens when we have more than one attribute?
- In the example before, we only had to consider annual income
- But what would happen if we had multiple attributes, like 5 or 10 or 100?
- **How do we decide which attribute to use?**

Decision Trees!

(now!)

Trees & Decisions Trees

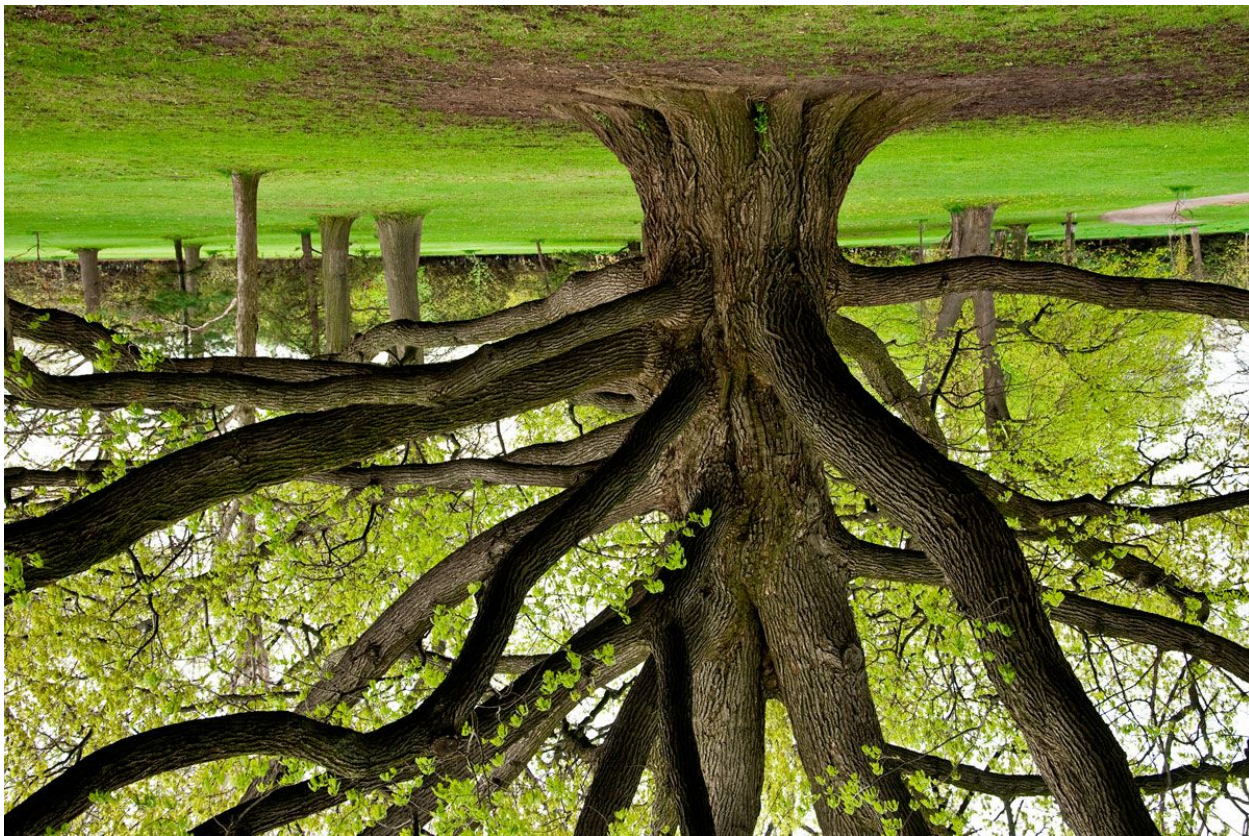
Regular Trees



Trees in Computer Science



Regular Trees

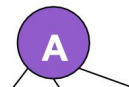


Trees in Computer Science

- A Decision Tree is a way for a computer to make decisions based on a series of questions.

Trees in Computer Science

- A Decision Tree is a way for a computer to make decisions based on a series of questions.



A **tree** is a **collection of nodes** such that

- One node is the designated ***root***.

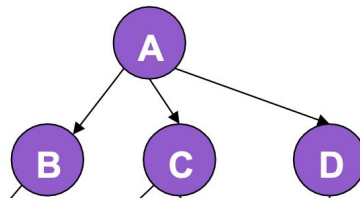
A is the root

Trees in Computer Science

- A Decision Tree is a way for a computer to make decisions based on a series of questions.

A **tree** is a **collection of nodes** such that

- One node is the designated ***root***.
- A node can have zero or more *children*;



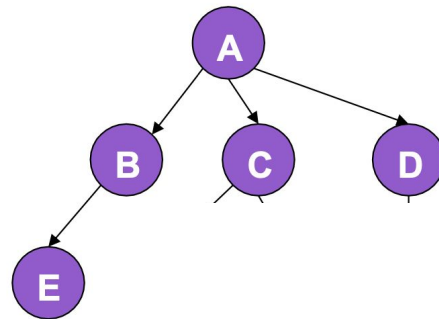
B, C and D are A's children

Trees in Computer Science

- A Decision Tree is a way for a computer to make decisions based on a series of questions.

A **tree** is a **collection of nodes** such that

- One node is the designated **root**.
- A node can have zero or more children;
- a node with zero children is a leaf.



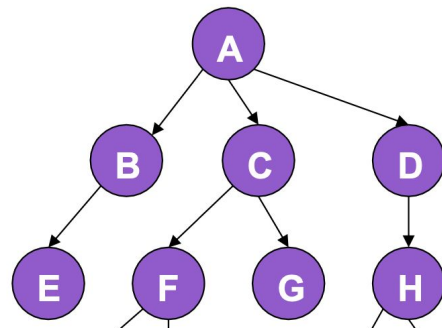
E is a leaf

Trees in Computer Science

- A Decision Tree is a way for a computer to make decisions based on a series of questions.

A **tree** is a **collection of nodes** such that

- One node is the designated **root**.
- A node can have zero or more children;
- a node with zero children is a leaf.
- All non-root nodes have a single parent.



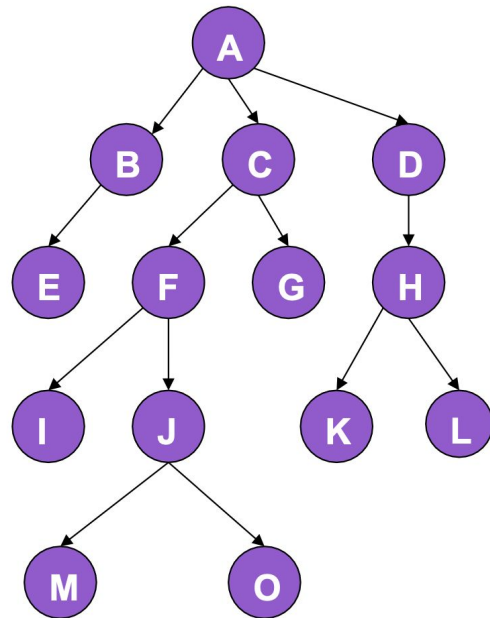
C is a parent to F + G

Trees in Computer Science

- A Decision Tree is a way for a computer to make decisions based on a series of questions.

A **tree** is a **collection of nodes** such that

- One node is the designated **root**.
- A node can have zero or more children;
- a node with zero children is a leaf.
- All non-root nodes have a single parent.
- Edges denote parent-child relationships.
 - Example: The arrows between $F \rightarrow I$

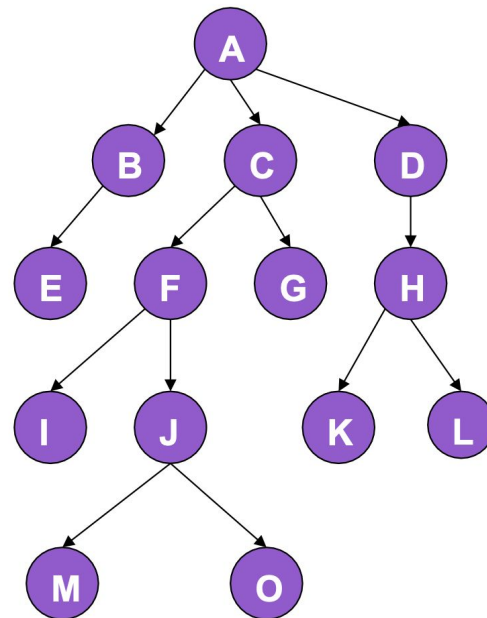


Trees in Computer Science

- A Decision Tree is a way for a computer to make decisions based on a series of questions.

A **tree** is a **collection of nodes** such that

- One node is the designated **root**.
- A node can have zero or more children;
- a node with zero children is a leaf.
- All non-root nodes have a single parent.
- Edges denote parent-child relationships.
- Nodes and/or edges may be labeled by data.
 - Each node on this tree is labeled by a letter



Learning Goals



Learning Goals

After this lecture, you should be able to:

- Explain the concept of a **rooted tree** and **decision tree**.
- Describe what the general decisions are in building a decision tree.
 - **Build a decision tree using entropy.**
- Describe **what considerations** are important in building a decision tree.

Clicker Questions (using Agora)



Clicker Question



Q: What is a classifier?

- A. *This option is intentionally left blank*
- B. A method to predict the future
- C. An algorithm that maps input data to a specific category
- D. A type of decision tree used for data mining
- E. A type of data storage for algorithms



Clicker Question



Q: What is a classifier?

A. *This option is intentionally left blank*

B. A method to predict the future

C. An algorithm that maps input data to a specific category

D. A type of decision tree used for data mining

E. A type of data storage for algorithms



Clicker Question

Q: In classification, how is the accuracy of a classifier evaluated?



- A. By comparing training data with random data
- B. By matching the classifier's results with decisions from test data
- C. By ensuring the classifier can handle large datasets
- D. By improving the efficiency of the algorithm



Clicker Question



Q: In classification, how is the accuracy of a classifier evaluated?

A. By comparing training data with random data

B. By matching the classifier's results with decisions from test data

C. By ensuring the classifier can handle large datasets

D. By improving the efficiency of the algorithm

Let's build a Decision Tree



Building Decision Trees

- Should you get an ice cream?
- You might start out with the following data

Weather	Wallet	Ice Cream?
Great	Empty	No
Nasty	Empty	No
Great	Full	Yes
Okay	Full	Yes
Nasty	Full	No



Building Decision Trees

- Should you get an ice cream?
- You might start out with the following data

Attributes

Conditions

Weather	Wallet	Ice Cream?
Great	Empty	No
Nasty	Empty	No
Great	Full	Yes
Okay	Full	Yes
Nasty	Full	No

Ice Cream Decision Tree



Should you get an ice cream?

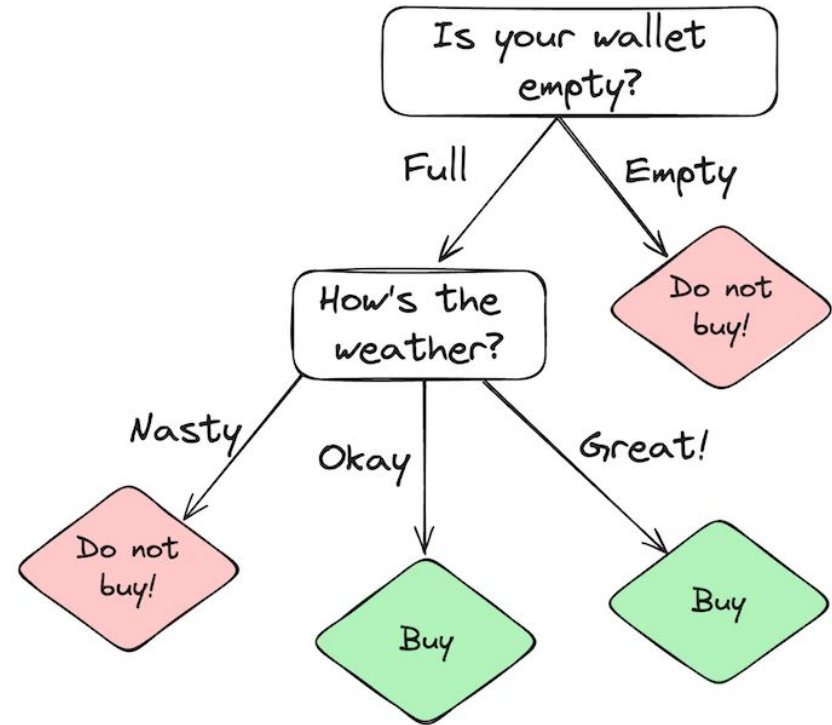
Weather	Wallet	Ice Cream?
Great	Empty	No
Nasty	Empty	No
Great	Full	Yes
Okay	Full	Yes
Nasty	Full	No



Should you get an ice cream?

Weather	Wallet	Ice Cream?
Great	Empty	No
Nasty	Empty	No
Great	Full	Yes
Okay	Full	Yes
Nasty	Full	No

Should I buy Ice Cream?



In-class Activity



In-class Activity: Decision Tree

Draw the decision tree with the data on the right for 13 people that are on the job market.

There is data on the salary as well the commute time...

#	Salary (\$)	Commute Time	Decision
1	120,000	45 min	Accept
2	105,000	10 min	Accept
3	100,000	25 min	Accept
4	90,000	20 min	Accept
5	75,000	15 min	Accept
6	65,000	60 min	Reject
7	55,000	20 min	Reject
8	50,000	25 min	Reject
9	45,000	70 min	Reject
10	30,000	40 min	Reject
11	95,000	25 min	Accept
12	80,000	10 min	Accept
13	110,000	50 min	Accept



**What if it is
not so clear ?**

Soccer League:

Do we cancel the game?



Soccer League: Cancel Game?

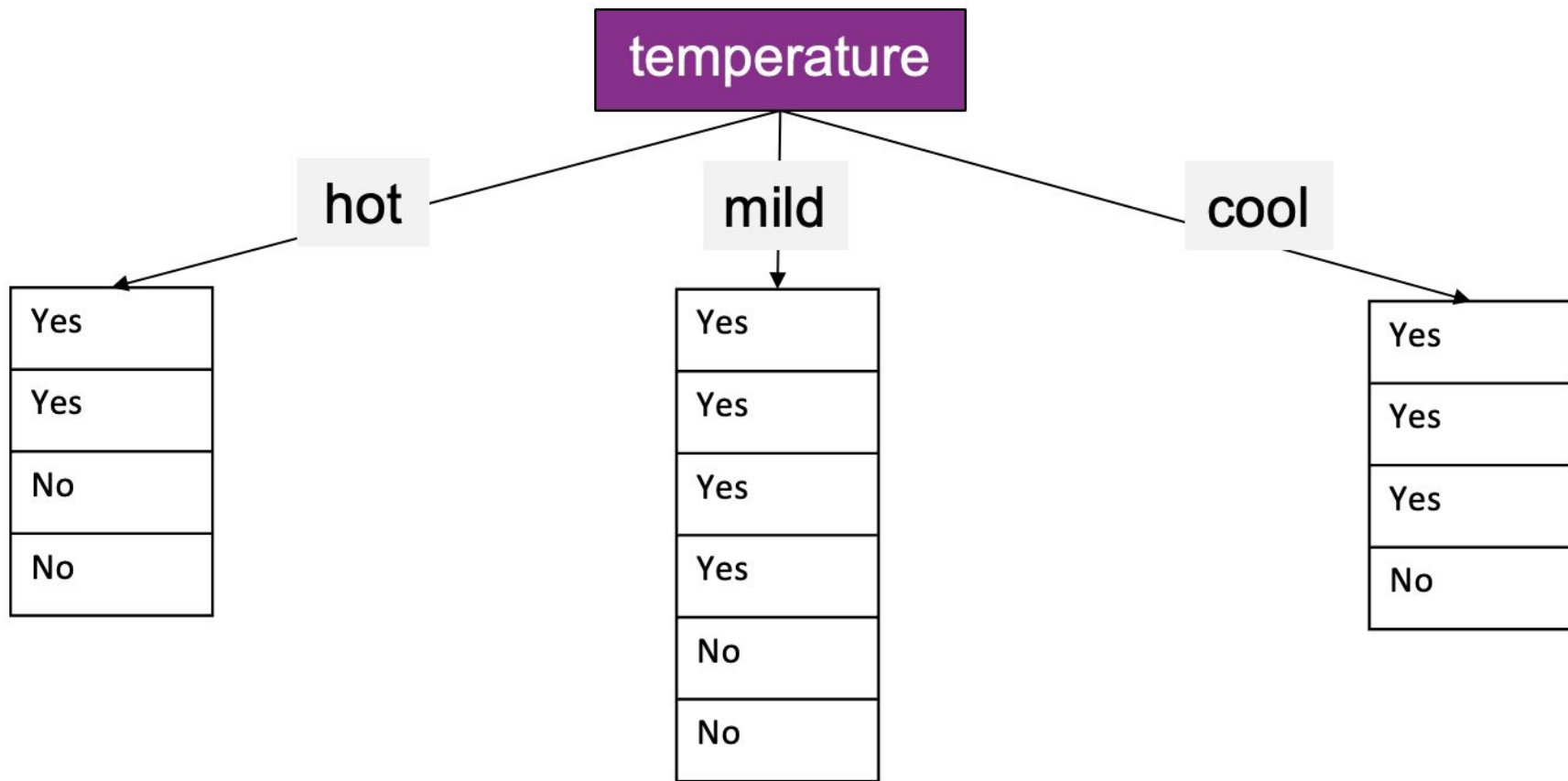
- Build a **decision tree** to help officials decide
- Assume that decisions are the same given the same information
- The leaf nodes should be whether or not to play
- The non-leaf nodes should be **attributes** (e.g., Outlook, Windy)
- The edges should be **conditions** (e.g., sunny, hot, normal)

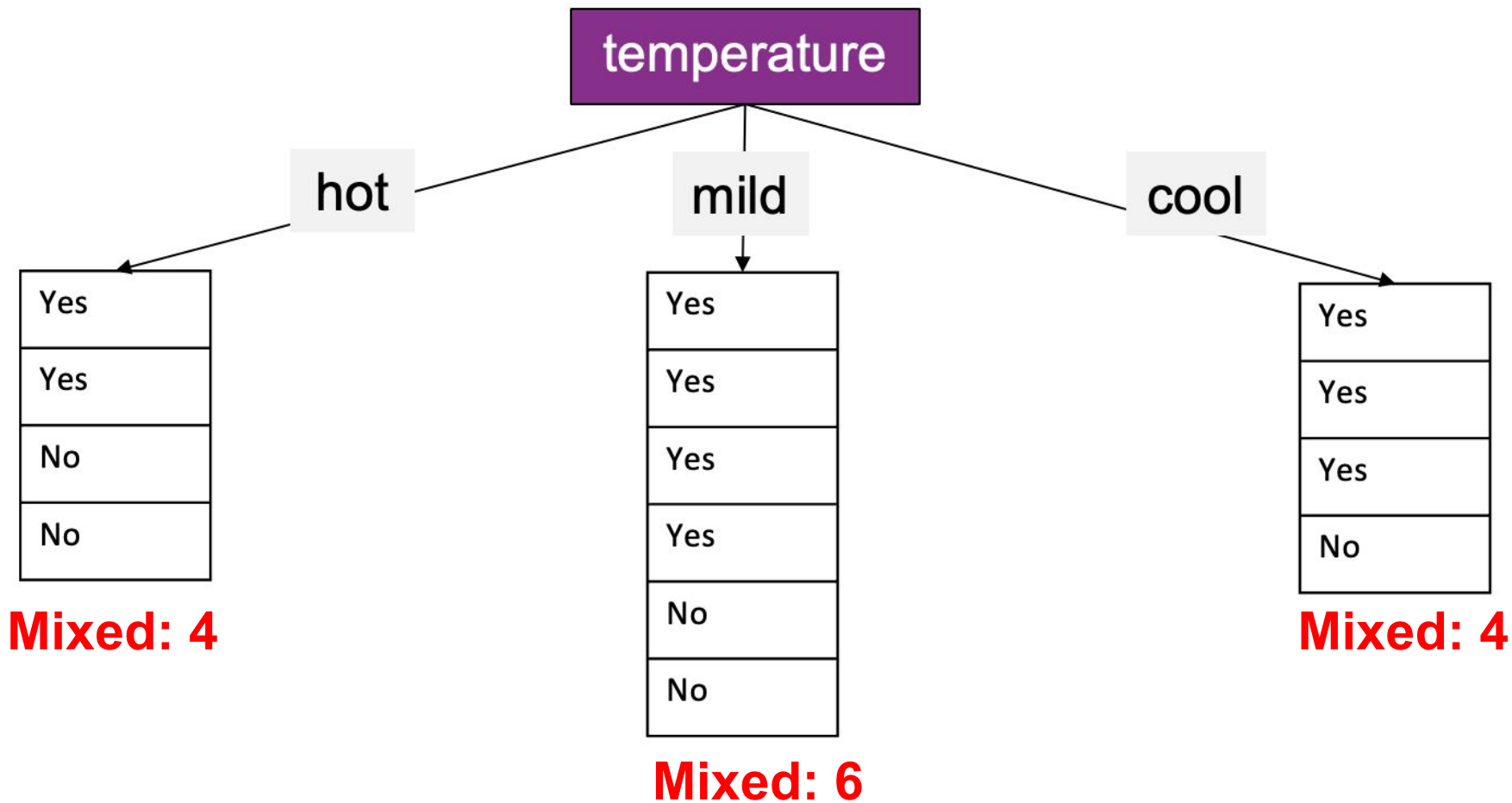
Want to have as few mixed “Yes” and “No” answers together in groups as possible.

At the start we have 14 mixed Yes’s/No’s

Outlook	Temperature	Humidity	Windy	Play?
sunny	hot	high	false	No
sunny	hot	high	true	No
overcast	hot	high	false	Yes
rain	mild	high	false	Yes
rain	cool	normal	false	Yes
rain	cool	normal	true	No
overcast	cool	normal	true	Yes
sunny	mild	high	false	No
sunny	cool	normal	false	Yes
rain	mild	normal	false	Yes
sunny	mild	normal	true	Yes
overcast	mild	high	true	Yes
overcast	hot	normal	false	Yes
rain	mild	high	true	No

**What happens if
we split data on
Temperature?**





What is the
uncertainty
(entropy) in our
data?

Overall entropy = 4 + 4 + 6 = 14

temperature

hot

mild

cool

Yes
Yes
No
No

Mixed: 4

Yes
Yes
Yes
Yes
No
No

Mixed: 6

Yes
Yes
Yes
No

Mixed: 4

Overall entropy = 4 + 4 + 6 = 14

temperature

hot

mild

cool

Yes
Yes
No
No

Mixed: 4

Yes
Yes
Yes
Yes
No
No

Mixed: 6

Note: The entropy for cool would be 0 if all of them were Yes or all of them were No (we are using simple math)

Yes
Yes
Yes
No

Mixed: 4

In-class Activity

[Groups of 3-4]

**What's the
entropy if you
split on the
Outlook?**

Outlook	Temperature	Humidity	Windy	Play?
sunny	hot	high	false	No
sunny	hot	high	true	No
overcast	hot	high	false	Yes
rain	mild	high	false	Yes
rain	cool	normal	false	Yes
rain	cool	normal	true	No
overcast	cool	normal	true	Yes
sunny	mild	high	false	No
sunny	cool	normal	false	Yes
rain	mild	normal	false	Yes
sunny	mild	normal	true	Yes
overcast	mild	high	true	Yes
overcast	hot	normal	false	Yes
rain	mild	high	true	No

Clicker Question



Q: What's the entropy if you split on Outlook?

A. 0

B. 5

C. 10

D. 14

E. None of the above

Outlook	Temperature	Humidity	Windy	Play?
sunny	hot	high	false	No
sunny	hot	high	true	No
overcast	hot	high	false	Yes
rain	mild	high	false	Yes
rain	cool	normal	false	Yes
rain	cool	normal	true	No
overcast	cool	normal	true	Yes
sunny	mild	high	false	No
sunny	cool	normal	false	Yes
rain	mild	normal	false	Yes
sunny	mild	normal	true	Yes
overcast	mild	high	true	Yes
overcast	hot	normal	false	Yes
rain	mild	high	true	No

Clicker Question



Q: What's the entropy if you split on Outlook?

A. 0

B. 5

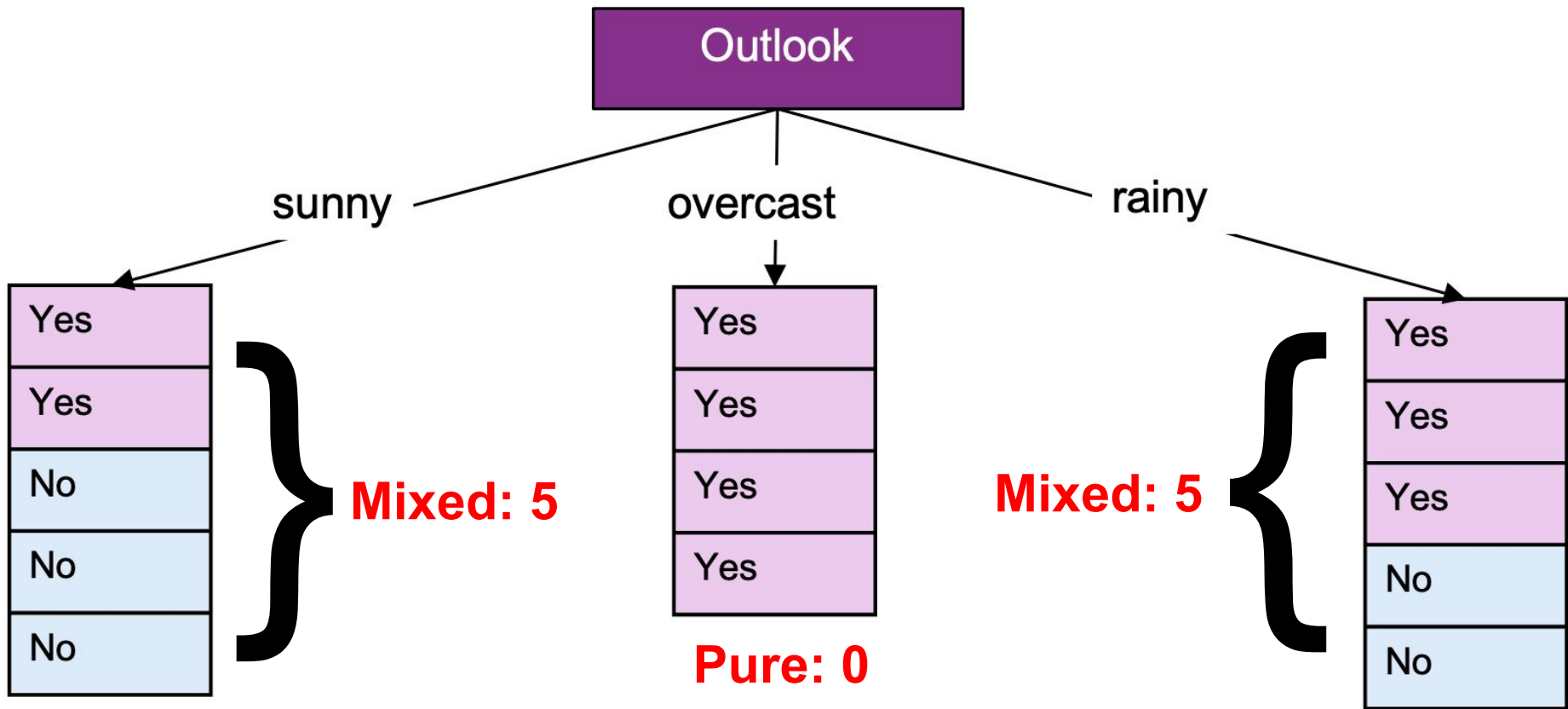
C. 10

D. 14

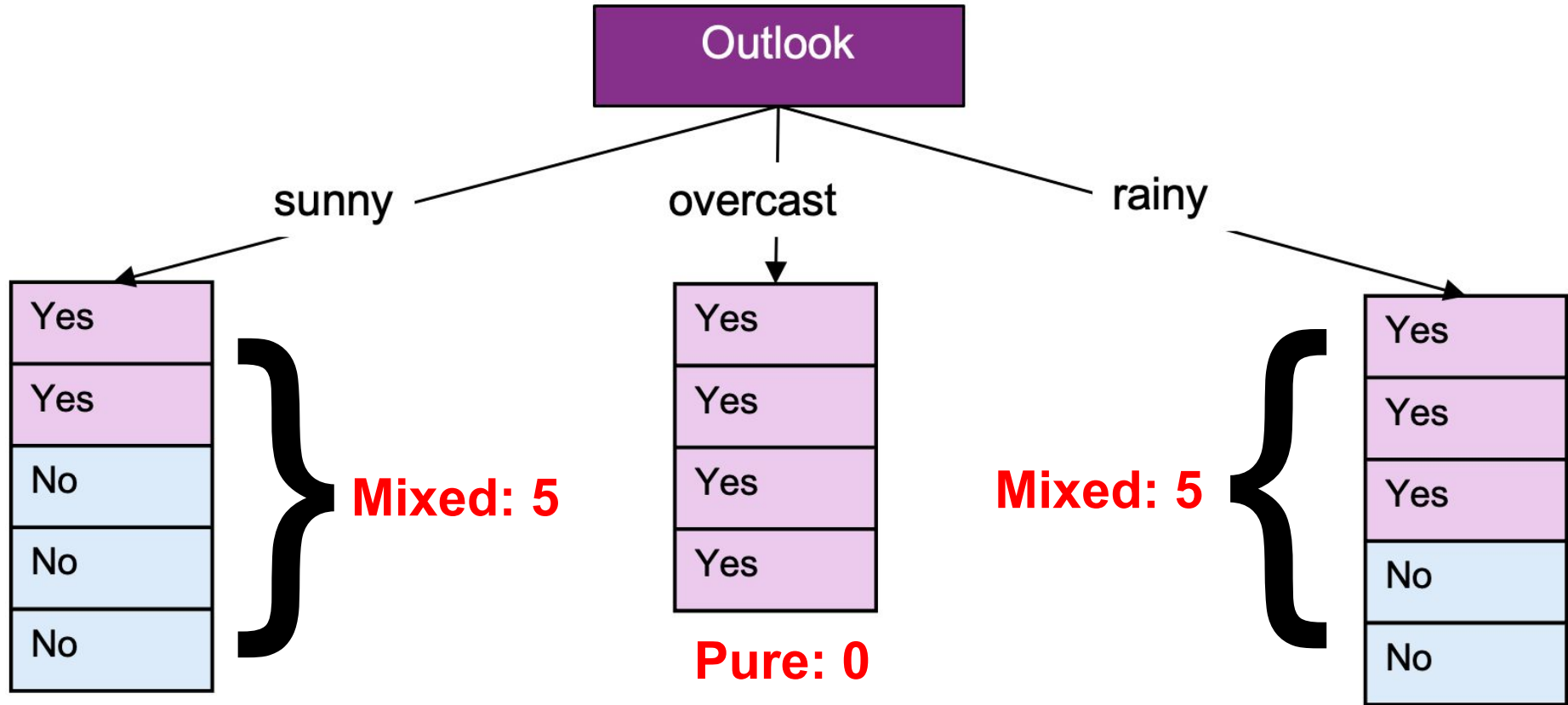
E. None of the above

Outlook	Temperature	Humidity	Windy	Play?
sunny	hot	high	false	No
sunny	hot	high	true	No
overcast	hot	high	false	Yes
rain	mild	high	false	Yes
rain	cool	normal	false	Yes
rain	cool	normal	true	No
overcast	cool	normal	true	Yes
sunny	mild	high	false	No
sunny	cool	normal	false	Yes
rain	mild	normal	false	Yes
sunny	mild	normal	true	Yes
overcast	mild	high	true	Yes
overcast	hot	normal	false	Yes
rain	mild	high	true	No

Q: What's the entropy if you split on Outlook?



$$\text{Overall entropy} = 5 + 0 + 5 = 10$$



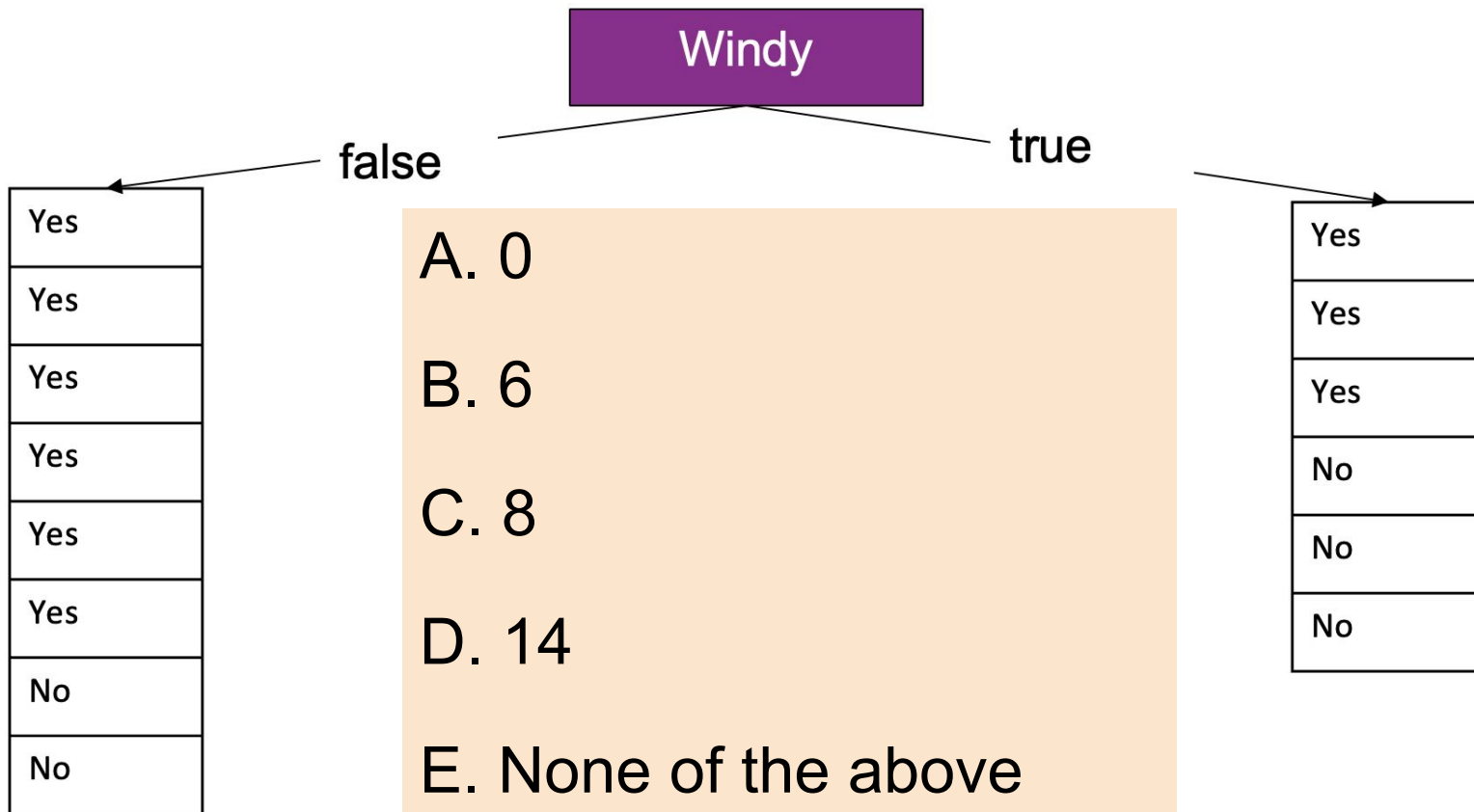
What's the entropy if you split on Windy?

Outlook	Temperature	Humidity	Windy	Play?
sunny	hot	high	false	No
sunny	hot	high	true	No
overcast	hot	high	false	Yes
rain	mild	high	false	Yes
rain	cool	normal	false	Yes
rain	cool	normal	true	No
overcast	cool	normal	true	Yes
sunny	mild	high	false	No
sunny	cool	normal	false	Yes
rain	mild	normal	false	Yes
sunny	mild	normal	true	Yes
overcast	mild	high	true	Yes
overcast	hot	normal	false	Yes
rain	mild	high	true	No



Clicker Question

Q: What's the entropy if you split on Windy?

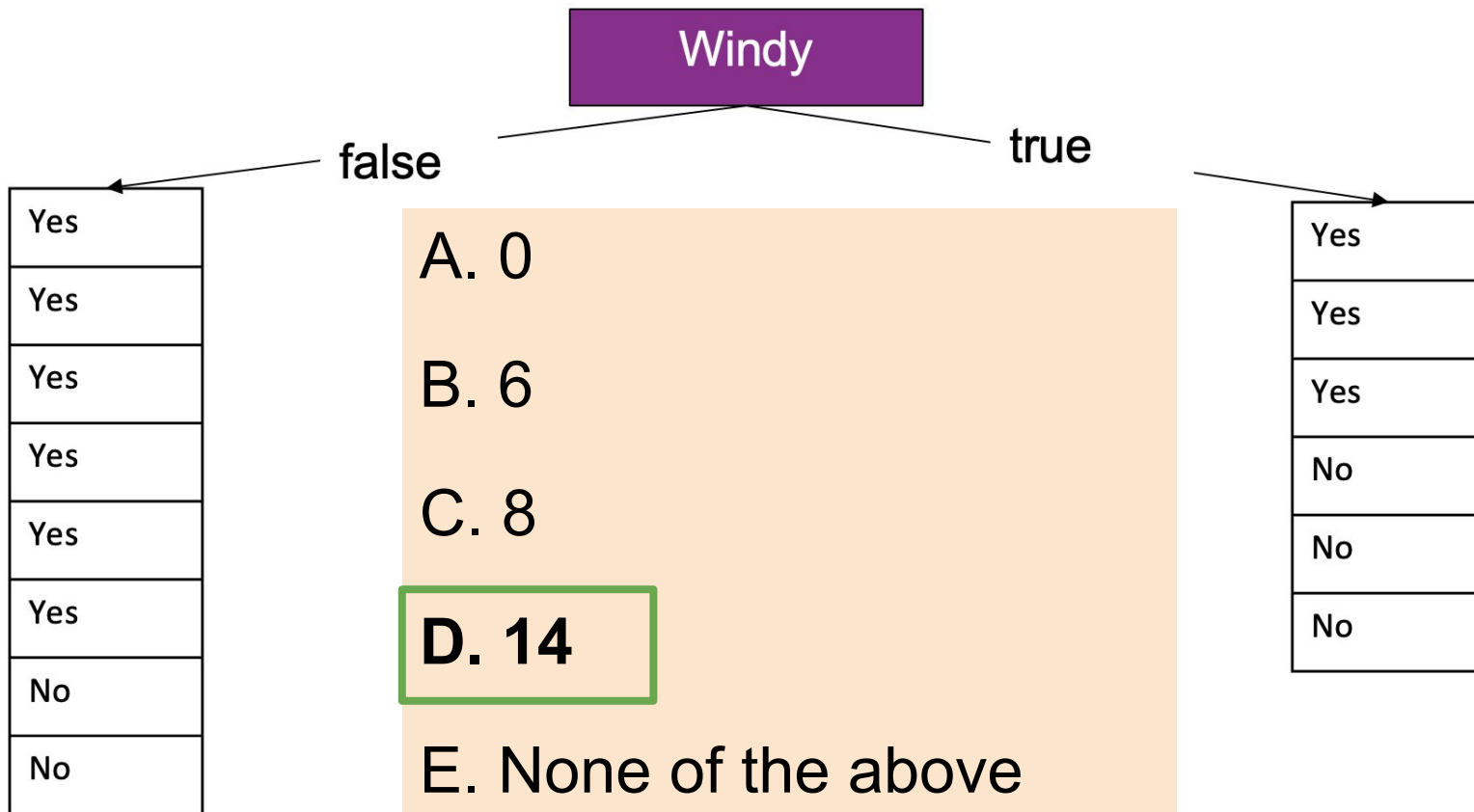


Clicker Question

Q: What's the entropy if you split on Windy?



iClicker

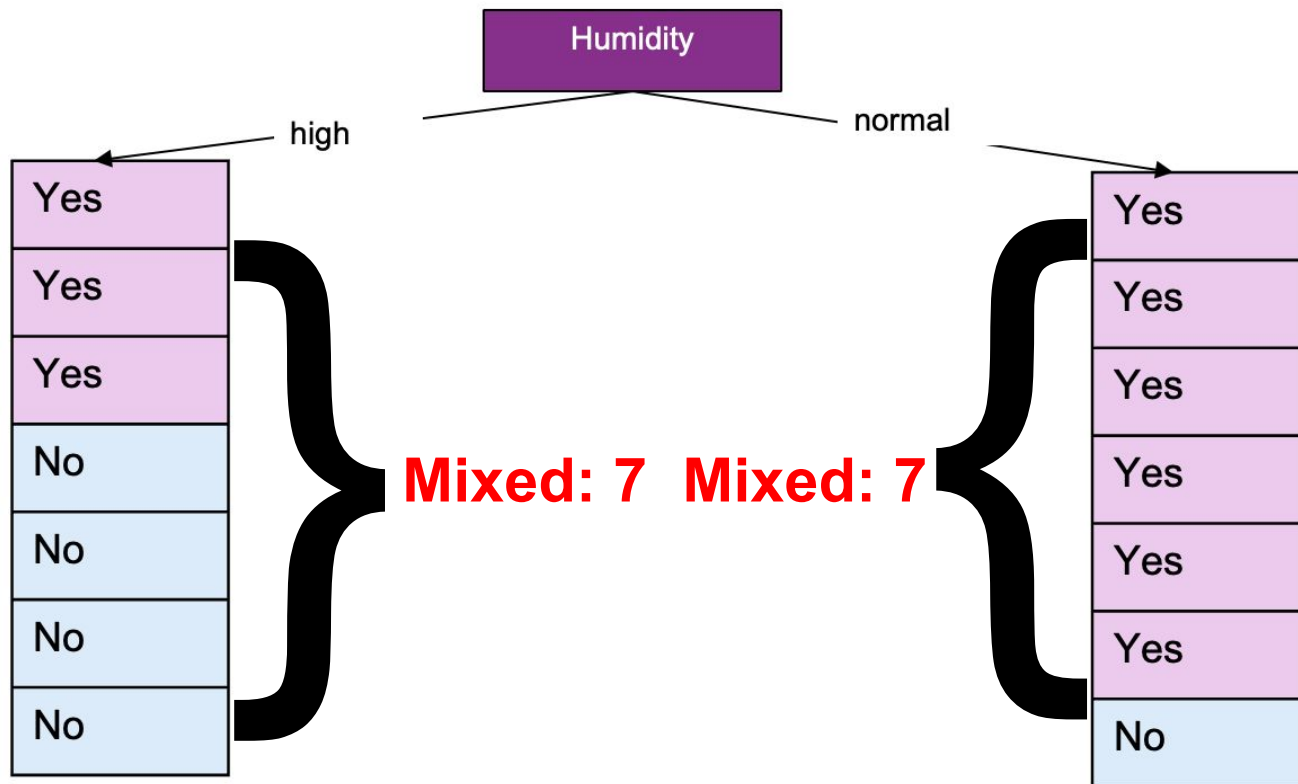


What's the entropy if you split on Humidity?

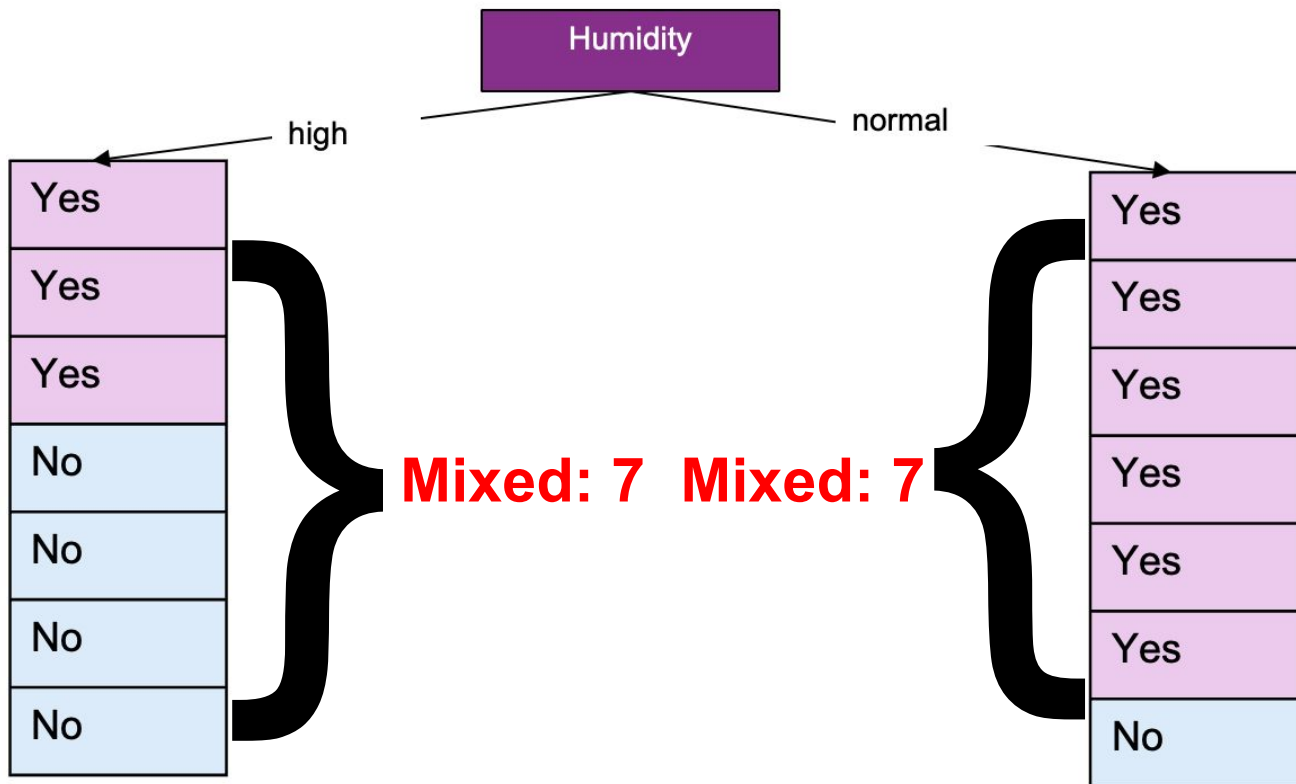
Outlook	Temperature	Humidity	Windy	Play?
sunny	hot	high	false	No
sunny	hot	high	true	No
overcast	hot	high	false	Yes
rain	mild	high	false	Yes
rain	cool	normal	false	Yes
rain	cool	normal	true	No
overcast	cool	normal	true	Yes
sunny	mild	high	false	No
sunny	cool	normal	false	Yes
rain	mild	normal	false	Yes
sunny	mild	normal	true	Yes
overcast	mild	high	true	Yes
overcast	hot	normal	false	Yes
rain	mild	high	true	No



Q: What's the entropy if you split on Humidity?



Overall entropy = 7 + 7 = 14



Debrief



What is the best attribute to split on?

- Entropy if we split on Temperature = 14
- Entropy if we split on Outlook = 10
- Entropy if we split on Windy = 14
- Entropy if we split on Humidity = 14

Why?



What is the best attribute to split on?

- Entropy if we split on Temperature = 14
- **Entropy if we split on Outlook = 10**
- Entropy if we split on Windy = 14
- Entropy if we split on Humidity = 14

Why? It does the best job of **minimizing** entropy

Wrap up